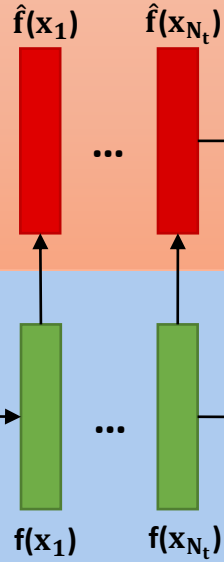
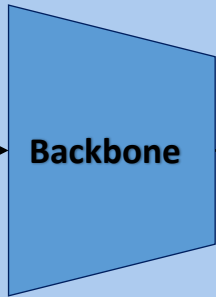
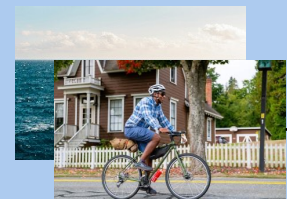
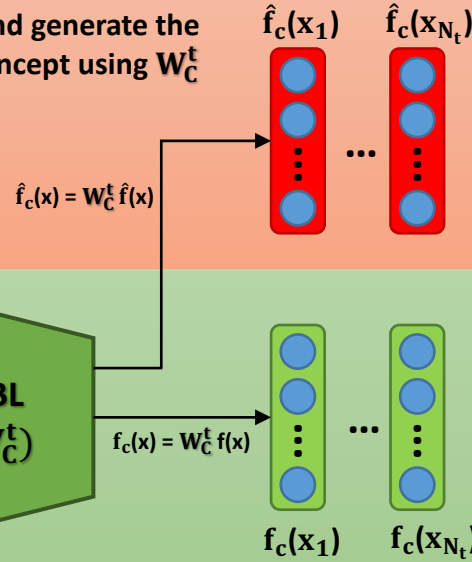


Step 2: Compute the embedding $f(x)$ using the frozen backbone and the concept matrix P_t based on X_t

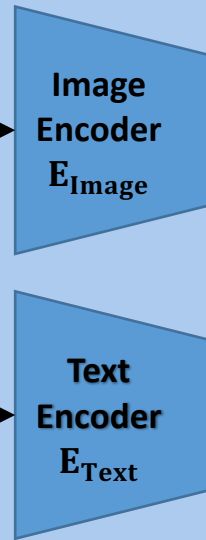
$X_t = \{x_1, \dots, x_{N_t}\}$



Step 4: Compute pseudo-features and generate the pseudo-concept using W_C^t

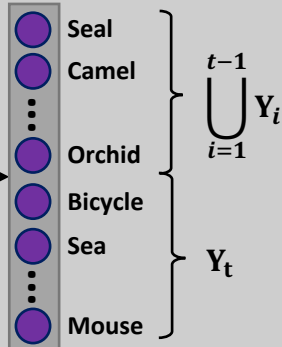
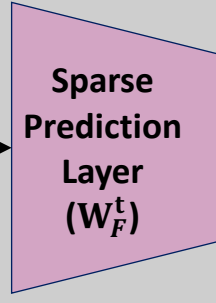


Step 3: Expand the CBL layer for new concepts and learn W_C^t by maximizing similarity and knowledge distillation.



$E_I(x_1) \cdot E_T(c_1)$	...	$E_I(x_1) \cdot E_T(c_{M_t})$
$E_I(x_2) \cdot E_T(c_1)$	...	$E_I(x_2) \cdot E_T(c_{M_t})$
...	...	...
$E_I(x_{N_t}) \cdot E_T(c_1)$	...	$E_I(x_{N_t}) \cdot E_T(c_{M_t})$

Step 5: Expand the prediction layer for new concepts and classes, and learn a sparse W_F^t using actual and pseudo-concepts.



Step1: Concept generation on new classes and unique concept set expansion

Bicycle
Sea
...
Mouse

GPT 3

C_{t-1}

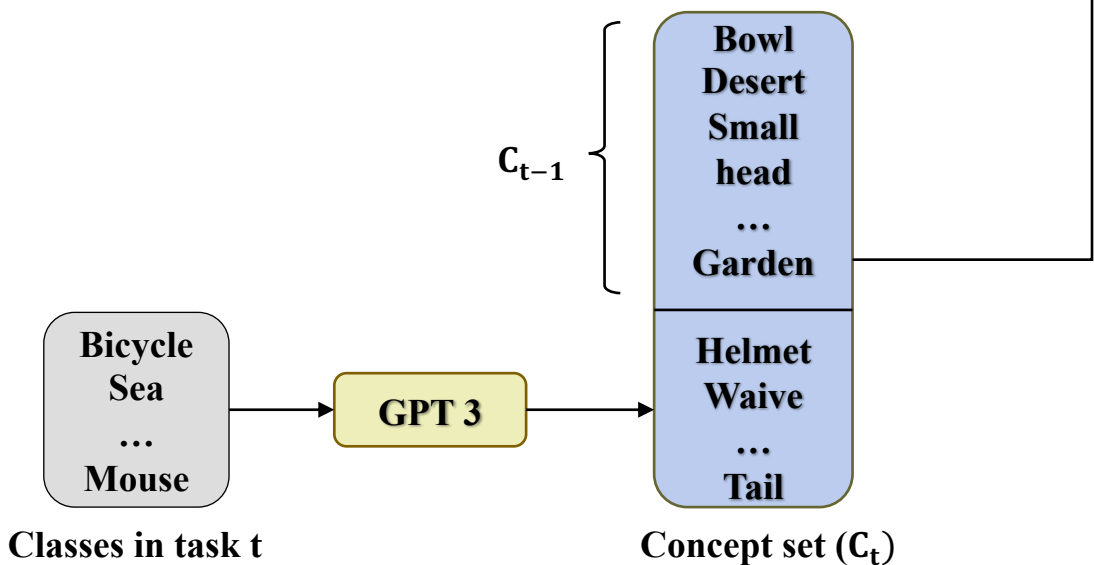
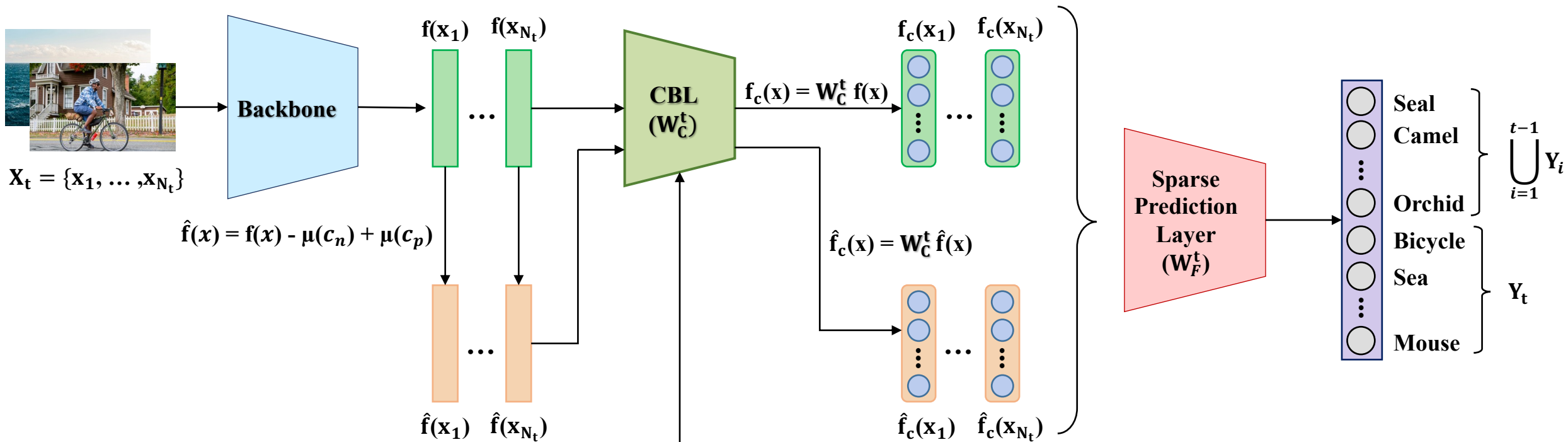
Bowl
Desert
A small head
...
Garden

Helmet
Waive
...
Long tail

Concept set (C_t)

P^t

Classes in task t



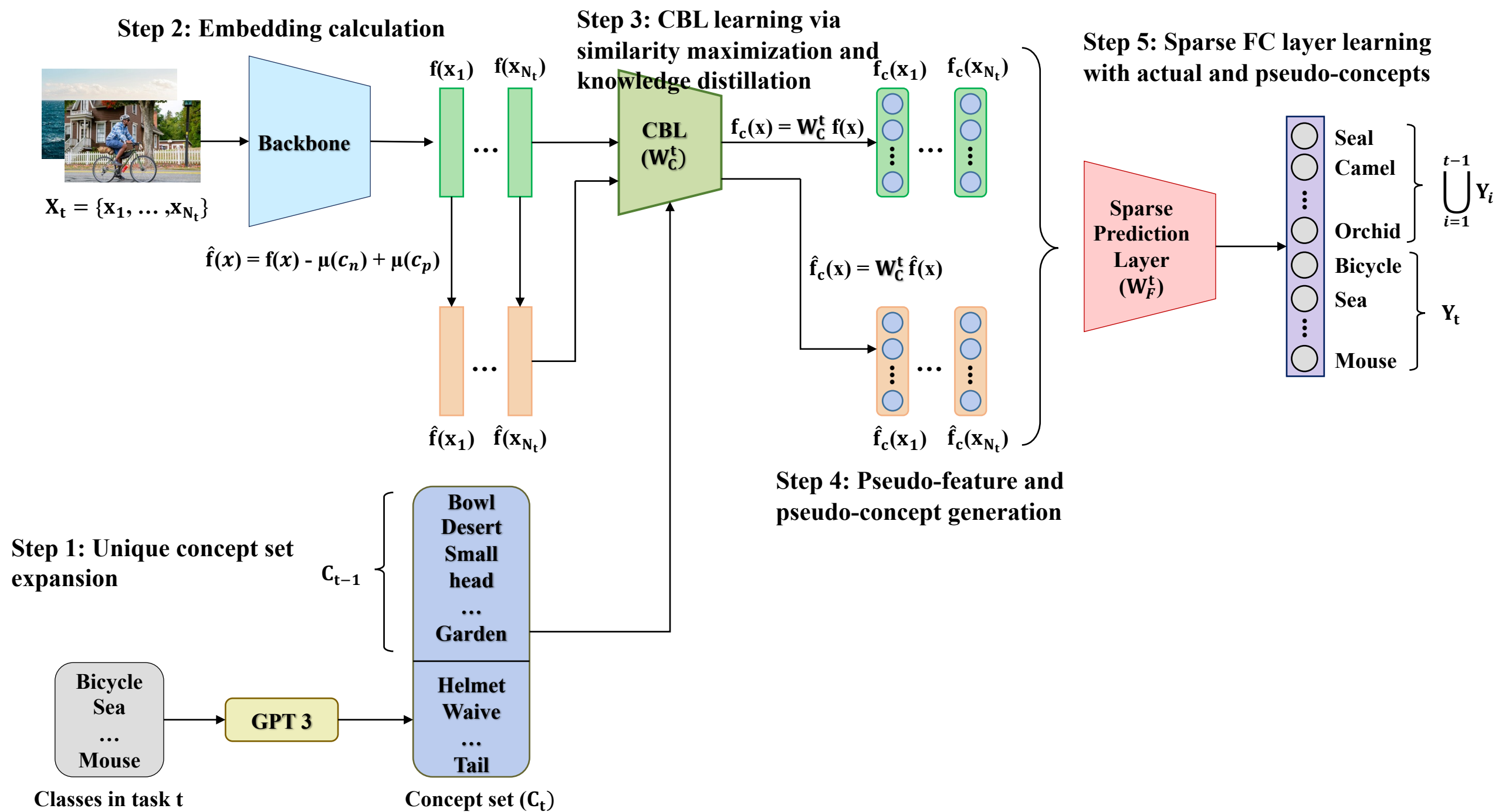
Step1: Concept generation on new classes and unique concept set expansion

Step 2: Compute the embedding $f(x)$ using the frozen backbone and the concept matrix P^t based on X_t

Step 3: Expand the CBL layer for new concepts and learn W_C^t by maximizing similarity and knowledge distillation.

Step 4: Compute pseudo-features and generate the pseudo-concept using W_C^t

Step 5: Expand the prediction layer for new concepts and classes, and learn a sparse W_F^t using actual and pseudo-concepts.



Step 1: Unique concept set expansion

C_{t-1}

Bowl
Desert
Small Head
...
Garden

Helmet
Waive
...
Tail

Concept set (C_t)

Bicycle
Sea
...
Mouse

GPT 3

Classes in task t

Step 3: CBL (W_C^t) learning via similarity maximization and knowledge distillation

Notation:

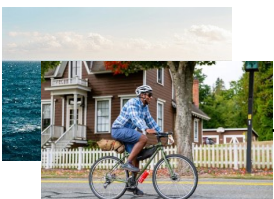
$$\hat{f}(x) = f(x) - \mu(c_n) + \mu(c_p)$$

$$\mu(c_k) = \frac{1}{N_k} \sum_{i=1}^{N_k} f(x_{k,i})$$

$$f_c(x) = W_C^t f(x)$$

$$\hat{f}_c(x) = W_C^t \hat{f}(x)$$

Step 5: Sparse FC layer (W_F^t) learning with actual and pseudo-concepts



$X_t = \{x_1, \dots, x_{N_t}\}$

Backbone

$f(x_1)$ $f(x_{N_t})$

Step 2: Embedding calculation based on X_t

CBL
(W_C^t)

$f_c(x_1)$ $f_c(x_{N_t})$

Step 4: Pseudo-feature and pseudo-concept generation

$\hat{f}(x_1)$ $\hat{f}(x_{N_t})$

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

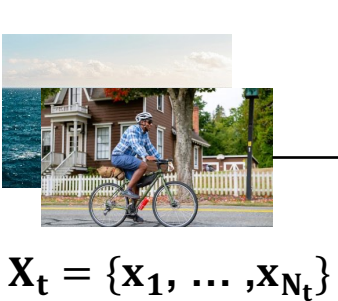
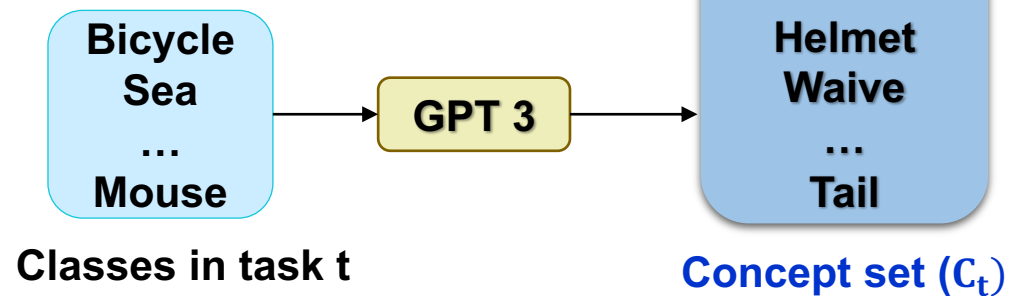
Sparse
Prediction
Layer
(W_F^t)

Seal
Camel
...
Orchid
Bicycle
Sea
...
Mouse

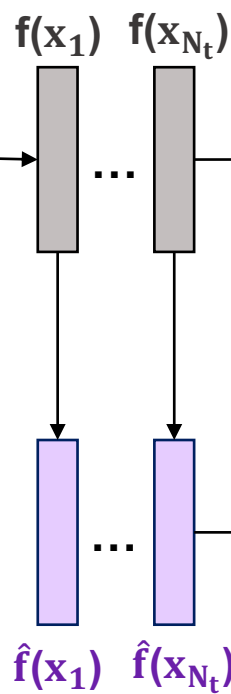
$\bigcup_{i=1}^{t-1} Y_i$

Y_t

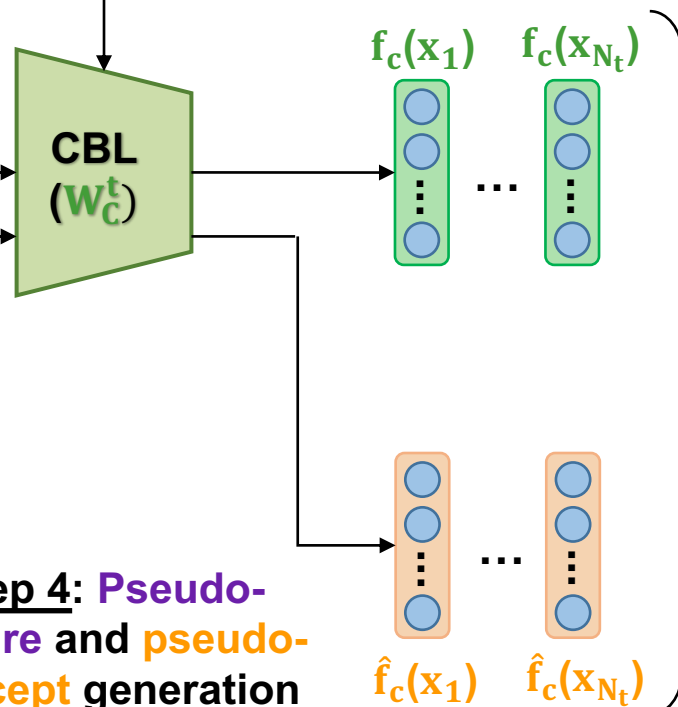
Step 1: Unique concept set expansion



Step 2: Embedding calculation based on X_t



Step 4: Pseudo-feature and pseudo-concept generation



Step 3: CBL (W_C^t) learning

Notation:

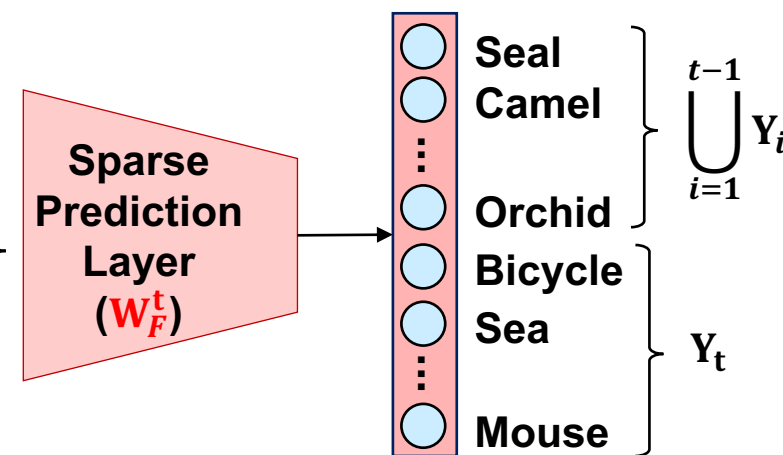
$$\hat{f}(x) = f(x) - \mu(c_n) + \mu(c_p)$$

$$\mu(c_k) = \frac{1}{N_k} \sum_{i=1}^{N_k} f(x_{k,i})$$

$$f_c(x) = W_C^t f(x)$$

$$\hat{f}_c(x) = W_C^t \hat{f}(x)$$

Step 5: Sparse FC layer (W_F^t) learning with actual and pseudo-concepts



I think using
arial font will probably
look better

use this kind of blue and make it light

use
light
blue
like this

**Step 1: Unique concept
set expansion**

Bicycle
Sea
...
Mouse

Classes in task t

GPT 3

C_{t-1}

Bowl
Desert
Small Head
...
Garden

Helmet
Waive
...
Tail

Concept set (C_t)

Step 3: CBL (W_C^t)

learning via similarity
maximization and
knowledge distillation

Notation:

$$\hat{f}(x) = f(x) - \mu(c_n) + \mu(c_p)$$

$$\mu(c_k) = \frac{1}{N_k} \sum_{i=1}^{N_k} f(x_{k,i})$$

$$f_c(x) = W_C^t f(x)$$

$$\hat{f}_c(x) = W_C^t \hat{f}(x)$$

we dotted
line.

make the
eqns aligned
left,
no
bold face

Step 5: Sparse FC layer (W_F^t)
learning with **actual** and
pseudo-concepts



$X_t = \{x_1, \dots, x_{N_t}\}$

Backbone

$f(x_1)$ $f(x_{N_t})$

...

$f_c(x_1)$ $f_c(x_{N_t})$

...

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

...

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

...

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

...

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

...

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

...

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

...

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

...

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

...

**Step 2: Embedding
calculation based on X_t**

**Step 4: Pseudo-
feature and pseudo-
concept generation**

Sparse
Prediction
Layer
(W_F^t)

Seal
Camel
...

Orchid
Bicycle
Sea
...

Mouse

$\bigcup_{i=1}^{t-1} Y_i$

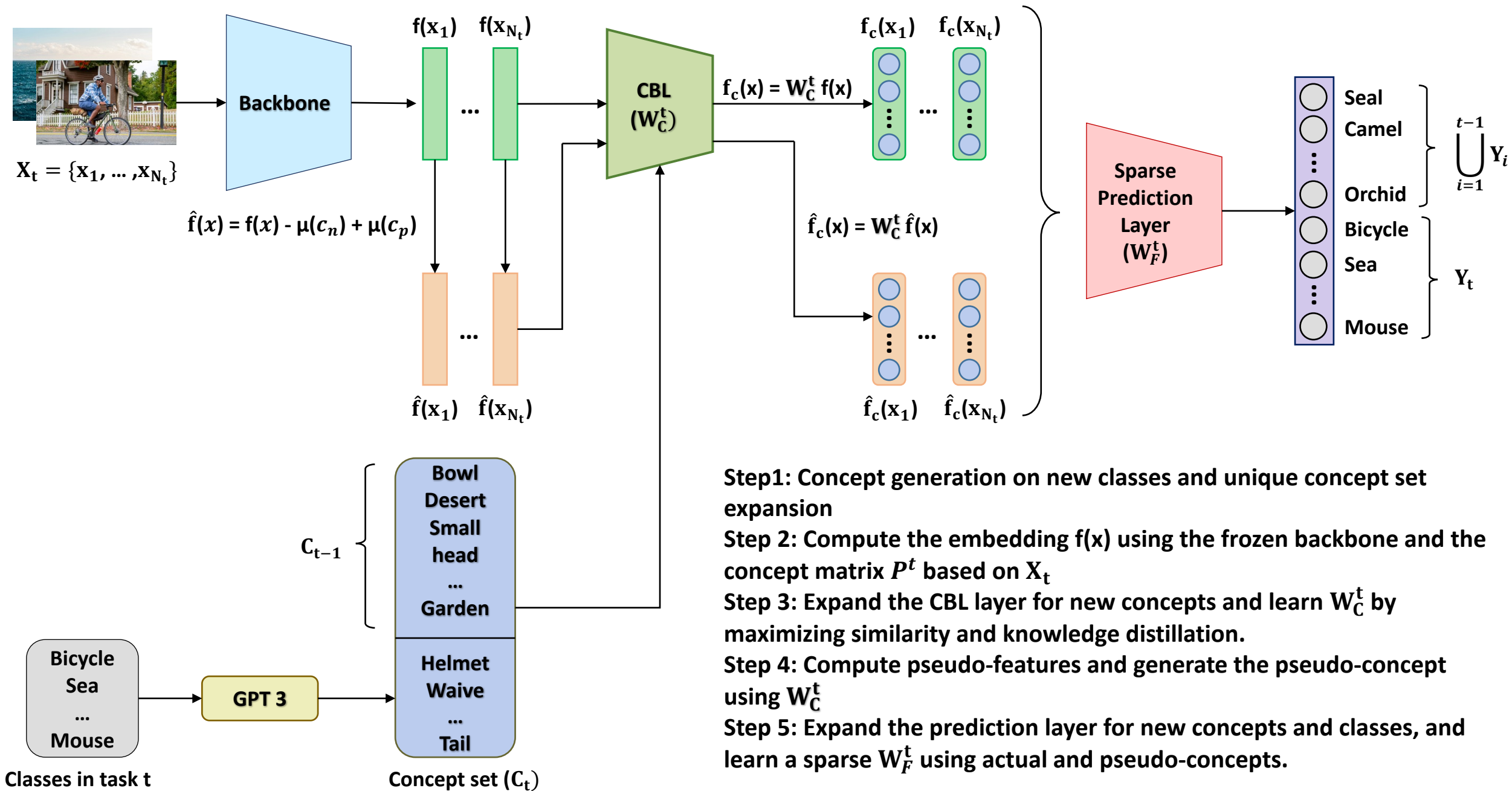
Y_t

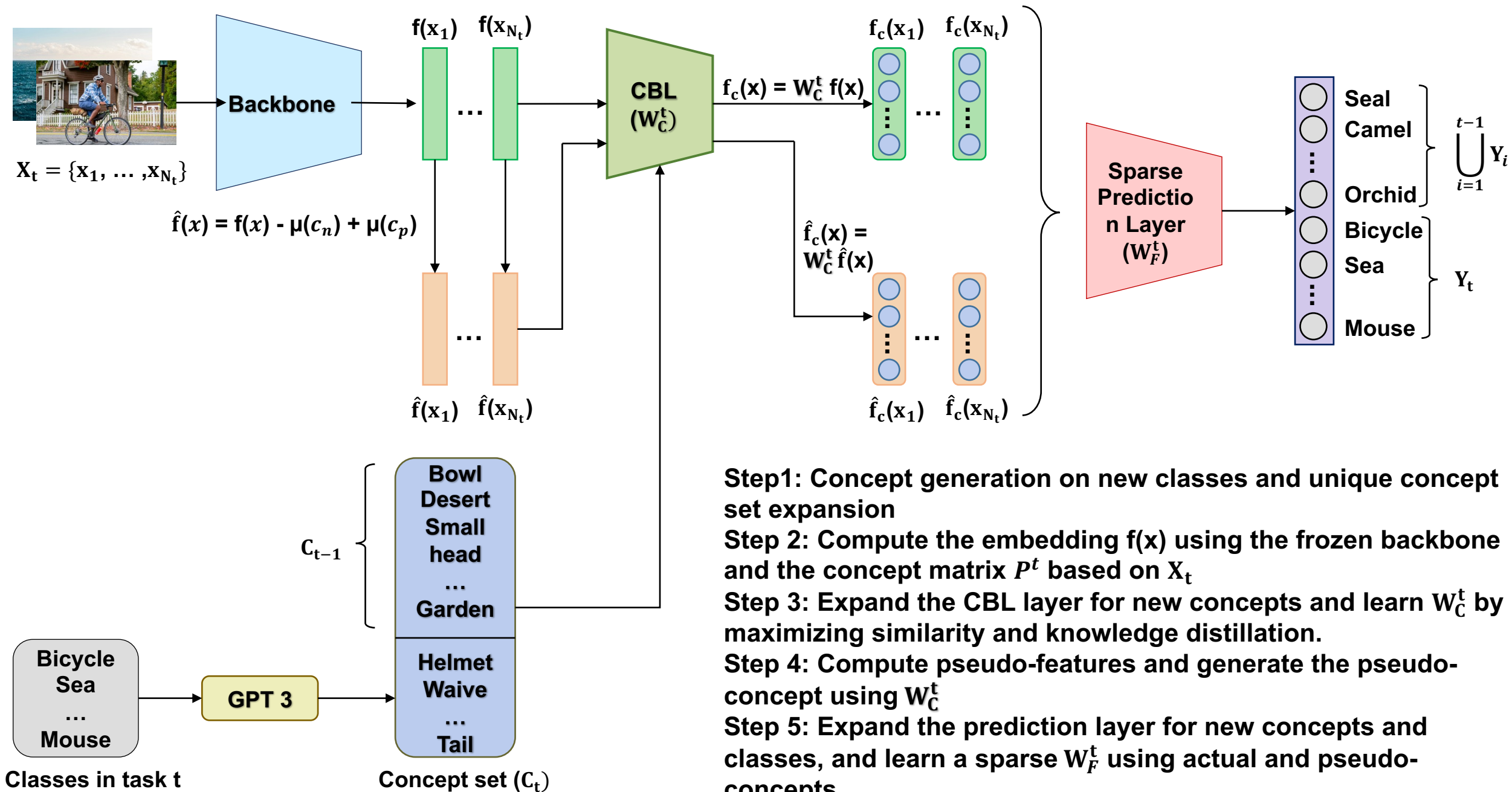
Figure 2. The architecture of CI-CBM

use
grey
color

use
light
grey

use grey color





step x:
make font
size larger
than other
font

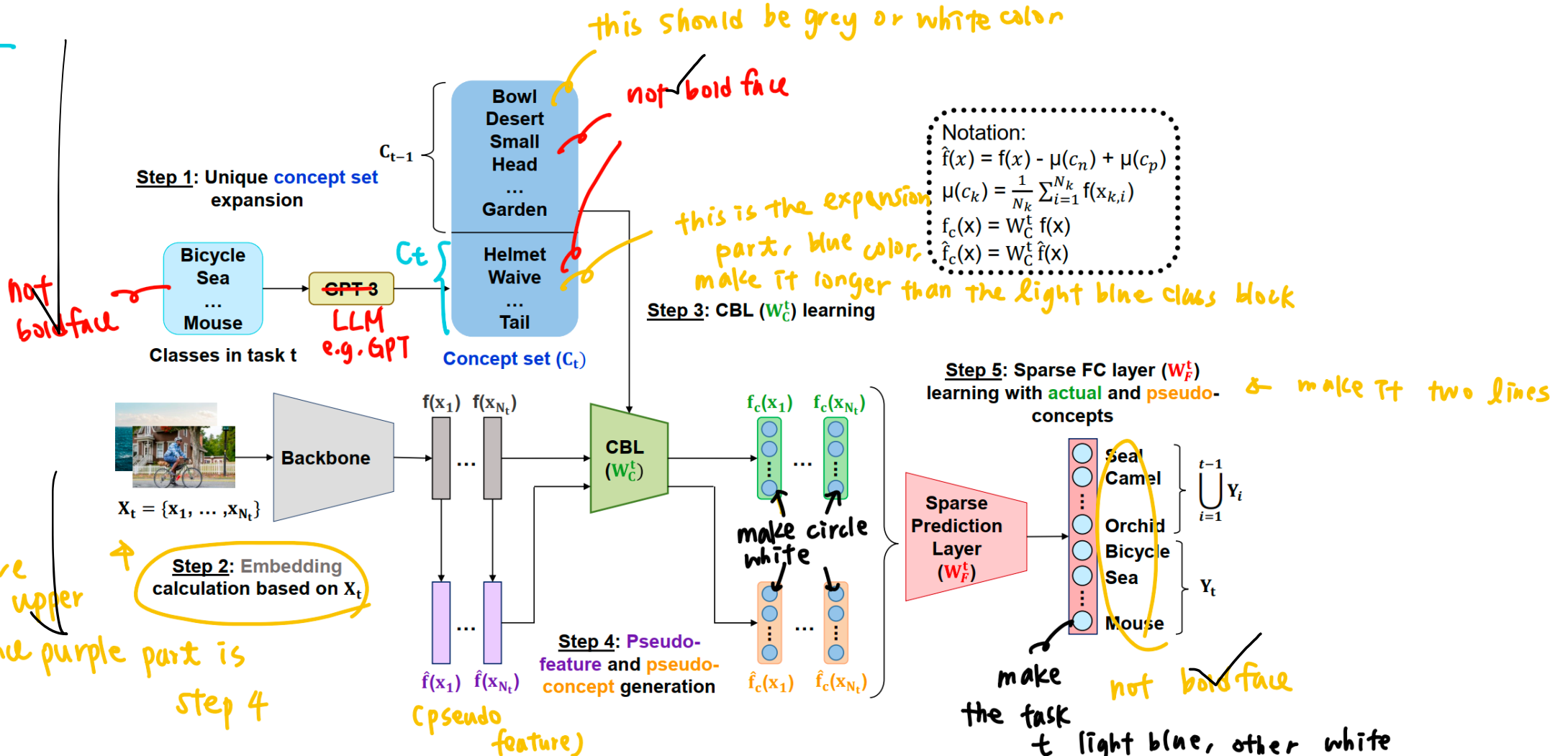


Figure 2: Overview of our pipeline for Class-Incremental CBM. Color-coded text matches the corresponding stages in the figure

Step 1: Unique concept set expansion

Bicycle
Sea
⋮
Mouse

LLM
e.g. GPT

C_{t-1}

Bowl
Sand
⋮
Garden
Helmet
Wheel
Wave
Blue
⋮
Tail

C_t

Concept Set (C_t)

Classes in task t

Notation:

$$\hat{f}(x) = f(x) - \mu(c_n) + \mu(c_p)$$

$$\mu(c_k) = \frac{1}{N_k} \sum_{i=1}^{N_k} f(x_{k,i})$$

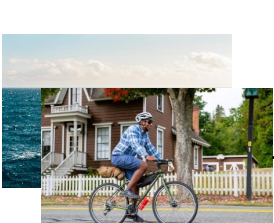
$$f_c(x) = W_C^t f(x)$$

$$\hat{f}_c(x) = W_C^t \hat{f}(x)$$

Step 3: CBL (W_C^t) learning

- Feature vector ($f(x)$)
- Pseudo-feature vector ($\hat{f}(x)$)
- Concept vector ($f_c(x)$)
- Pseudo-concept vector ($\hat{f}_c(x)$)
- Old concept unit from old tasks
- New concept unit for task t
- Class in old tasks
- Class in task t

Step 5: Sparse FC layer (W_F^t) learning with actual and pseudo-concepts



$X_t = \{x_1, \dots, x_{N_t}\}$

Backbone

$f(x_1)$ $f(x_{N_t})$

CBL
(W_C^t)

$f_c(x_1)$ $f_c(x_{N_t})$

$\hat{f}(x_1)$ $\hat{f}(x_{N_t})$

Step 4: Pseudo-feature and pseudo-concept generation

$\hat{f}_c(x_1)$ $\hat{f}_c(x_{N_t})$

Sparse Prediction Layer
(W_F^t)

Seal
Camel
⋮
Orchid
Bicycle
Sea
⋮
Mouse

$\bigcup_{i=1}^{t-1} Y_i$

Y_t